

Problem Set 5
Due: Mar. 29, 2021

1. Section 2.6 Problem 32.

D. Bernoulli's problem of joint lives. Consider $2m$ persons forming m couples who live together at a given time. Suppose that at some later time, the probability of each person being alive is p , independent of other persons. At that later time, let A be the number of persons that are alive and let S be the number of couples in which both partners are alive. For any survivor number a , find $\mathbf{E}[S \mid A = a]$.

Solution: Let X_i be the random variable taking the value 1 or 0 depending on whether the first partner of the i th couple has survived or not. Let Y_i be the corresponding random variable for the second partner of the i th couple. Then, we have $S = \sum_{i=1}^m X_i Y_i$, and by using the total expectation theorem,

$$\begin{aligned}\mathbf{E}[S \mid A = a] &= \sum_{i=1}^m \mathbf{E}[X_i Y_i \mid A = a] \\ &= m \mathbf{E}[X_1 Y_1 \mid A = a] \\ &= m \mathbf{E}[Y_1 \mid X_1 = 1, A = a] P(X_1 = 1 \mid A = a) \\ &= m P(Y_1 = 1 \mid X_1 = 1, A = a) P(X_1 = 1 \mid A = a).\end{aligned}$$

We have

$$P(Y_1 = 1 \mid X_1 = 1, A = a) = \frac{a-1}{2m-1}, \quad P(X_1 = 1 \mid A = a) = \frac{a}{2m}.$$

Thus,

$$\mathbf{E}[S \mid A = a] = m \frac{a-1}{2m-1} \cdot \frac{a}{2m} = \frac{a(a-1)}{2(2m-1)}.$$

Note that $\mathbf{E}[S \mid A = a]$ does not depend on p .

2. Section 2.7 Problem 42.

Computational problem. Here is a probabilistic method for computing the area of a given subset S of the unit square. The method uses a sequence of independent random selections of points in the unit square $[0, 1] \times [0, 1]$, according to a uniform probability law. If the i th point belongs to the subset S the value of a random variable X_i is set to 1, and otherwise it is set to 0. Let $X_1, X_2, \dots$ be the sequence of random variables thus defined, and for any n , let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- (a) Show that $\mathbf{E}[S_n]$ is equal to the area of the subset S , and that $\text{var}(S_n)$ diminishes to 0 as n increases.

- (b) Show that to calculate S_n , it is sufficient to know S_{n-1} and X_n , so the past values of X_k , $k = 1, \dots, n-1$, do not need to be remembered. Give a formula.
- (c) Write a computer program to generate S_n for $n = 1, 2, \dots, 10000$, using the computer's random number generator, for the case where the subset S is the circle inscribed within the unit square. How can you use your program to measure experimentally the value of π ?
- (d) Use a similar computer program to calculate approximately the area of the set of all (x, y) that lie within the unit square and satisfy $0 \leq \cos \pi x + \sin \pi y \leq 1$.

Solution:

- (a) Noting that

$$P(X_i = 1) = \frac{\text{Area}(S)}{\text{Area}([0, 1] \times [0, 1])} = \text{Area}(S),$$

we obtain

$$\mathbb{E}[S_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \text{Area}(S),$$

and

$$\text{var}(S_n) = \text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i) = \frac{1}{n} \text{var}(X_i) = \frac{1}{n}(1 - \text{Area}(S))\text{Area}(S),$$

which tends to be zero as n tends to infinity.

- (b) We have

$$S_n = \frac{n-1}{n}S_{n-1} + \frac{1}{n}X_n.$$

- (c) We can generate S_{10000} (up to a certain precision) as follows:

- i. Initialize S to zero.
- ii. For $i = 1$ to 10000
- iii. Randomly select two real numbers a and b (up to a certain precision) independently and uniformly from the interval $[0, 1]$.
- iv. If $(a - 0.5)^2 + (b - 0.5)^2 < 0.25$, set x to 1 else set x to 0.
- v. Set $S := (i-1)S/i + x/i$.
- vi. Return S .

By running the above algorithm, a value of S_{10000} equal to 0.7783 was obtained (the exact number depends on the random number generator). We know from part (a) that the variance of S_n tends to zero as n tends to infinity, so the obtained value of S_{10000} is an approximation of $\mathbb{E}[S_{10000}]$. But $\mathbb{E}[S_{10000}] = \text{Area}(S) = \pi/4$, this leads us to the following approximation of π :

$$4 \cdot 0.7783 = 3.1132.$$

- (d) We only need to modify the test done at step iv. We have to test whether or not $0 \leq \cos \pi a + \sin \pi b \leq 1$. The obtained approximation of the area was 0.3755.
3. Every package of some intrinsically dull commodity includes a small and exciting plastic object. There are c different types of object, and each package is equally likely to contain any given type. You buy one package each day.
- (a) Find the mean number of days which elapse between the acquisitions of the j th new type of object and the $(j + 1)$ th new type.
- (b) Find the mean number of days which elapse before you have a full set of objects.

Solution:

- (a) If you have already j distinct types of object, the probability that the next packet contains a different type is $(c - j)/c$, and the probability that it does not is j/c . Hence the number of days required has the geometric distribution with parameter $(c - j)/c$; this distribution has mean $c/(c - j)$.
- (b) The time required to collect all the types is the sum of the successive times to collect each new type. The mean is therefore
- $$\sum_{j=0}^{c-1} \frac{c}{c-j} = c \sum_{k=1}^c \frac{1}{k}.$$
4. Each trial may result in any of t given outcomes, the i th outcome having probability p_i . Let N_i be the number of occurrences of the i th outcome in n independent trials.

- (a) Show that

$$P(N_i = n_i \text{ for } 1 \leq i \leq t) = \frac{n!}{n_1! n_2! \cdots n_t!} p_1^{n_1} p_2^{n_2} \cdots p_t^{n_t}$$

for any collection of $n_1, n_2, \dots, n_t$ of non-negative integers with sum n . The vector N is said to have the multinomial distribution.

- (b) Find the marginal mass functions of the multinomial distribution.

Solution:

- (a) There are $n!/(n_1! n_2! \cdots n_t!)$ sequences of outcomes in which the i th possible outcome occurs n_i times for each i . The probability of any such sequence is $p_1^{n_1} p_2^{n_2} \cdots p_t^{n_t}$, and the result follows.
- (b) Let N_i be the number of times the i th outcome occurs. Then N_i has the binomial distribution with parameters n and p_i .